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Export diversification and economic growth in sub-Saharan Africa: could there be a mixed relationship?

Jocelyne Urielle BEDIME
Faculty of Economics and Applied Management University of Douala- Cameroon Research Group in Economics and Management (GREG), bedimejoycy@gmail.com

Julien Brice MINKANDE
Faculty of Economics and Applied Management University of Douala - Cameroon Research Group in Economics and Management (GREG)

Honoré BIDIASSE
Faculty of Economics and Applied Management University of Douala - Cameroon Research Group in Economics and Management (GREG

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ORIGINAL ARTICLE

Export Diversification and Economic Growth in Sub-Saharan Africa: Could There be a Mixed Relationship?

Jocelyne Urielle Bedime*, Julien Brice Minkande, Honoré Bidiasse

Research Group in Economics and Management (GREG), Faculty of Economics and Applied Management, University of Douala, P.O. Box 4032, Douala, Cameroon

Abstract

Based on the differences in export diversification between sub-regions, this research aims to determine the type of relationship that may exist between export diversification and economic growth in Sub-Saharan Africa. Using data from 36 countries grouped according to their level of diversification, a panel ARDL model is estimated. The results show that for highly diversified sub-regions, the relationship between export diversification and economic growth is inverted U-shaped, while for concentrated sub-regions, the relationship is U-shaped. Highly diversified countries are advised to integrate further into the African continental free-trade zone, which, by lifting barriers and improving the regulatory framework, offers greater openness to new markets. For less diversified countries, it would be wise to accumulate commodity rents, as this will enable them to bear the costs of implementing diversification policies when the time comes.

Keywords: Export diversification, Economic growth, SSA, ARDL, Sub-regions, PMG

1. Introduction

In recent years, export diversification has gained renewed interest in the macroeconomic literature following the negative effects of the financial and health crises that have affected the world's economies (Djampin & Bita, 2023; Yuni et al., 2020). These negative effects were particularly severe for developing economies, especially those in Sub-Saharan Africa (SSA) (Djampin & Bita, 2023). The main argument put forward to justify these negative impacts has been the lack of export diversification in these economies. A country that relies heavily on a single export product runs a high risk of suffering economic shocks (Mania & Rieber, 2019; Yuni et al., 2020). To this end, export diversification is recommended, as it not only helps maintain export earnings despite price volatility on world markets but above all stimulates economic growth at the same time (Gutierrez De Piñeres & Ferrantino, 2000; Lederman & Maloney, 2012).

In the literature, the consensus on the benefits of export diversification has two main sources (Mania & Rieber, 2019). On the one hand, the empirical research conducted by Imbs and Wacziarg (2003) and, on the other, the successful industrialization experience of Asian newly industrializing countries (NICs) (Mania & Rieber, 2019). Indeed, three theoretical justifications explain the importance of export diversification on economic growth. First, export diversification helps reduce export instability, considered to be one of the causes of slow growth rates and highly volatile exchange rates (Herzer & Nowak-Lehmann, 2006). Second, export diversification generates new production technologies and managerial efficiencies through international competition, leading to increasing returns to scale and spillover effects that promote growth and

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* Corresponding author.
E-mail address: bedimejoycy@gmail.com (J. Urielle Bedime).

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development (Hodey et al., 2015). Finally, diversification appears to facilitate "catch-up" effects between countries (Linder, 1961; Posner, 1961; Prebisch, 1950; Singer, 1950). This is because it enables the commercial and production structures of developing countries to tend toward those of industrialized countries (Siswana & Phiri, 2020).

Over the years, numerous studies have analyzed the effects of export diversification on economic growth at transnational, regional, and national levels. These studies reveal a range of somewhat divergent results. While some authors find that export diversification positively affects countries' economic growth (Agosin, 2007; Matadeen, 2011; Sannassee et al., 2014), others do not (Ferreria & Harrison, 2012; Lee & Zhang, 2019; McIntyre et al., 2018). While the first two groups of authors presuppose a linear relationship, a third hypothesizes an inverted U-shaped non-monotonic relationship (Imbs & Wacziarg, 2003; Klinger & Lederman, 2006; Naudé & Rossouw, 2011). According to this third hypothesis, to maintain economic growth, developing countries must first diversify their economies to create a more equitable balance between the different sectors of the economy. Once this stage of the development process has been reached, the nation must once again concentrate on its exports to grow further. Thus, diversification has a positive effect when the country is developing, and from a certain level of higher per capita income, the relationship tends to reverse (Klinger & Lederman, 2006; Naudé & Rossouw, 2011).

When we look specifically at the SSA region, we find that uncertainties remain as to the precise nature of the relationship. Three main results emerge: first, export diversification stimulates economic growth. This view is defended by Songwe and Winkler (2012), Hodey et al. (2015), Napo and Adjande (2019), and Yuni et al. (2020). Second, export diversification has no significant effect on economic growth (Mania & Rieber, 2019; Yokoyama & Alemu, 2009). The third point of view is that of Cabral and Veiga (2012), who believe that export diversification hurts economic growth. Here, the negative effect and lack of significant effect are mainly justified by the still low level of export diversification.

However, the level of export diversification reveals a certain heterogeneity between the different sub-regions (UNCTADSTAT, 2021). Indeed, while Central Africa and West Africa show low levels of diversification, Southern Africa and East Africa have more diversified export structures, with indices between 0.14 and 0.18 (UNCTADSTAT, 2021). Thus, beyond these contradictory results, we note that, despite the relatively large number of empirical works, almost no studies have examined the relationship by considering the difference in levels of export diversification that exist between sub-regions. Studies that have focused on groups of countries have mainly concentrated on the area-specific effect (Hodey et al., 2015) or income levels (Yuni et al., 2020). Yet, with the ideas of harmonization and consolidation of regional objectives materialized by the establishment of the African Continental Free Trade Area (ACFTA) to correct the unevenly paced progress of economic communities (CNUCED, 2019), an analysis based on diversification gaps could provide interesting new insights into the relationship between export diversification and economic growth, specifically between sub-regions and generally across SSA. What's more, in resource-exporting countries such as those in SSA, a high-income level is not always an indicator of the level of internal development of structures (Yokoyama & Alemu, 2009). Income levels are even less indicative of an economy's ability to cope with shocks (Hodey et al., 2015). On the other hand, diversification provides information on the process of structural transformation and therefore on a country's economic development (IMF 2014; McIntyre et al., 2018). Indeed, diversification makes it possible both to capitalize on current comparative advantage and to build the capacities needed to create new comparative advantages.

Thus, considering the differences in export diversification between sub-regions, the objective of the present study is to determine the type of relationship that exists between export diversification and economic growth in SSA. This research contributes to the existing literature by examining the specific functional relationship between export diversification and economic growth in SSA under the contradictory hypothesis of an inverted-U relationship, established by Imbs and Wacziarg (2003).

In this study, we use a panel ARDL model based on the Pooled Mean Group (PMG) estimator. This PMG estimator, developed by Pesaran et al. (1999), makes it possible to distinguish between the short and long term, considering country heterogeneity. For the rest, the work is presented as follows: Sect. 2 presents the literature review, Sect. 3 gives the

Abbreviations

ARDL	AutoRegressive Distributed Lag
PMG	Pooled Mean Group
CNUCED	Conférence des Nations Unies sur le Commerce et le Développement

methodology, and Sect. 4 presents the results and discusses them. Finally, conclusions and recommendations are made in Sect. 5.

2. Literature review

2.1. Theoretical review

Theoretically, the debate over the role of diversification in economic growth is not new; it was already at the heart of the controversies between structuralists and free-trade advocates in the 1950s. Based on Ricardo's conventional ideas of international trade (1817), free-trade economists promoted free trade and specialization based on nations' comparative advantages. The new theories of international trade, introduced by Krugman (1979), updated this result by arguing for the existence of increasing returns to scale. They demonstrate that when an economy is open to international trade, export concentration can lead to gains.

However, Prebisch (1950) and Singer (1950) challenged these ideas of specialization, arguing that in emerging countries, comparative advantage inevitably led to growth, but that this growth was modest. According to these economists, differences in economic structures between countries led to a trade pattern in which industrialized countries exported high-value-added industrial products, while developing countries exported primary or labor-intensive goods. However, exports from developing countries are characterized by low income elasticity, low productivity, and excessive price fluctuations. As a result, unstable export earnings and deteriorating terms of trade condemn developing economies to impoverishing growth. Export diversification is therefore recommended as a remedy because it reduces an economy's sensitivity to foreign demand shocks and promotes the technical spillovers necessary for economic growth (Mania & Rieber, 2019).

The modern structuralist school, for its part, has adopted the critique brought to comparative advantage theory by asserting that the diversification and composition of a developing economy's exports are the drivers of structural change (Botta, 2010; Cimoli et al., 2010, 2011; Vera, 2006). Consequently, development programs must introduce distortions into the specialization processes of comparative advantage theory (Lederman & Maloney, 2012). Indeed, many researchers use the fact that Chinese and other Asian NICs industrialized by challenging their comparative advantages and diversifying their export structure to support their theoretical arguments (Amsden, 1989; Lin & Chang, 2009; Rodrik, 2012; Wade, 1990).

Theoretical convergence has occurred, with new iterations of conventional models including export diversification as an engine of economic growth. These so-called "new trade theory" models (Chaney, 2008; Feenstra & Kee, 2008; Melitz, 2003) are based on an extension of Krugman's (1979) model and incorporate inter-firm productivity heterogeneity into international trade models. They demonstrate that an increase in the number of exporting companies results in a decrease in trade costs. Lower trade costs will result in greater export diversity, as each firm produces several different items under monopolistic competition. The redistribution of market share to the most productive firms leads to an improvement in overall productivity through a selection effect. These models show how trade intensification leads to an economy's export diversity, which in turn promotes economic growth.

Neostructuralists and conventional economists disagree, however. For the latter, export diversification occurs solely through lower trade costs, while for the former, it requires intentional state intervention. Beyond this important debate on economic policy, a consensus has developed around the idea that export diversification promotes economic growth and development.

2.2. Empirical review

Researchers agree that since the pioneering work of Imbs and Wacziarg (2003), the empirical literature has been enriched with works that study the link between export diversification and growth under the hypothesis of a nonlinear, U-shaped (or inverted U-shaped) relationship (Cadot et al., 2011; Siswana & Phiri, 2020; Yuni et al., 2020). It should be noted, however, that the empirical literature specific to SSA is still limited, even though export diversification is one of the countries' most desired objectives (Yuni et al., 2020). For their part, the few publications close to the present study diverge on the shape of this relationship. For example, Hodey et al. (2015), assuming a hump-shaped relationship, examine the link between export diversification and growth in 42 SSA countries. According to their results, export diversification has a positive effect on economic growth. However, they found no evidence of a nonlinear relationship between the two variables. Nevertheless, when comparing countries by subregion, they find that, compared to East Africa, the effect of export diversification on growth is higher for Central and Southern African countries. On the contrary, the effect of export diversification on growth is weaker in West Africa than in East Africa.

In the same spirit of research, Yuni et al. (2020) assessed the relationship between export diversification and economic growth in selected SSA countries. Using fixed-effects and generalized least-squares regression models, they make a comparative study between 39 countries classified according to their income level (low, medium, high). The results of their analysis reveal a U-shaped relationship between export concentration and economic growth. This result shows a non-significant positive relationship for low-income countries, a positive and significant relationship for lower-middle-income countries, and a negative but non-significant relationship for upper-middle-income countries. This leads the authors to believe that export diversification is advantageous at a particular stage of an economy's development.

In view of the above literature, there is no conclusive empirical evidence on the shape of the relationship between export diversification and economic growth in African countries. Although there is strong theoretical support that a diversified export structure has positive effects on economic growth, it is undeniable that empirical analysis can paint a different picture. To this end, one is always entitled to ask: do sub-regions with a more diversified export structure have consistently high growth? Or does the curve reverse at some point? What about sub-regions with a less diversified export structure?

3. Methodology

3.1. Sampling

To highlight the effects linked to the level of export diversification, we create two sub-samples corresponding to SSA sub-regions according to their levels of export diversification. The first sub-sample (E1) is made up of 17 countries from the most diversified sub-regions (Southern Africa and East Africa). The second (E2) is made up of 19 countries from the least diversified sub-regions (Central and West Africa). The list of countries by sub-sample is presented in Appendix 1. The grouping of sub-regions is based on Herfindhal-Hirschman concentration index data from the UNCTADstat database. The dividing line between sub-regions is chosen based on a concentration index ratio of 0.2 (Fig. 1a and b).

Indeed, on the one hand, if we look at Fig. 1a, which compares the concentration levels of the world's regions, we can see that, compared with the world's other developing regions (South Asia, Latin America, and the Caribbean and the developing economies), SSA has concentration levels above the 0.2 limit. On the other hand, referring to Fig. 1b, which shows the concentration levels of the SSA sub-regions, we note that Southern Africa and East Africa are below the 0.2 limit, while West and Central Africa are well above it. Considering that Southern Africa and East Africa have concentration levels equal to those of developing countries, it makes sense to study them separately from West and Central Africa.

3.2. Data and variables

The database used in this study covers annual time series for 36 SSA countries over the period 1995–2019. The choice of sample and analysis period is guided by the availability of data on the variables in the different databases used. The logarithm was applied to variables with very high values to reduce their scale. Data on real gross domestic product (GDP), human capital index (HUCAP), physical capital stock (PHYCAP), rate of depreciation of physical capital (DELTA), and employment level (EMPLOYMENT) come from Penn State Word Tables, and data on population growth rate (DEMO) come from World Development Indicators. Finally, the database of the United Nations Conference on Trade and Development (UNCTAD) was used to collect the data required to calculate the Theil export concentration index used in this study as an indicator of export diversification. The Theil index is calculated as follows:

$$\text{Theil} = \frac{1}{n}\sum_{k=1}^{n}\frac{x_k}{\mu}\ln\left(\frac{x_k}{\mu}\right) \tag{1}$$

with $\mu = \frac{1}{n}\sum_{k=1}^{n} x_k$, n being the number of export lines, and x_k being the export value of the product k.

The Theil index was chosen for its decomposition property. This property is its main asset. Unlike other concentration indices (Herfindahl-Hirschman index, Gini index), which capture a country's exports in aggregate terms (shares), the Theil index measures exports not only in terms of maintaining or increasing shares but also in terms of increasing the number of active lines or product lines (Bruckner, 2023). As diversification reflects the real evolution of a country's productive activities (Cadot et al., 2011), the Theil index provides a better understanding of a country's diversification policy orientations. It therefore helps to advise policymakers on how economic activity can be distributed between existing products/sectors and on the possibility of extending the country's export portfolio to new sectors (Cadot et al., 2011). The Theil index is also relatively insensitive to inter-temporal price changes, making it a better diversification measure

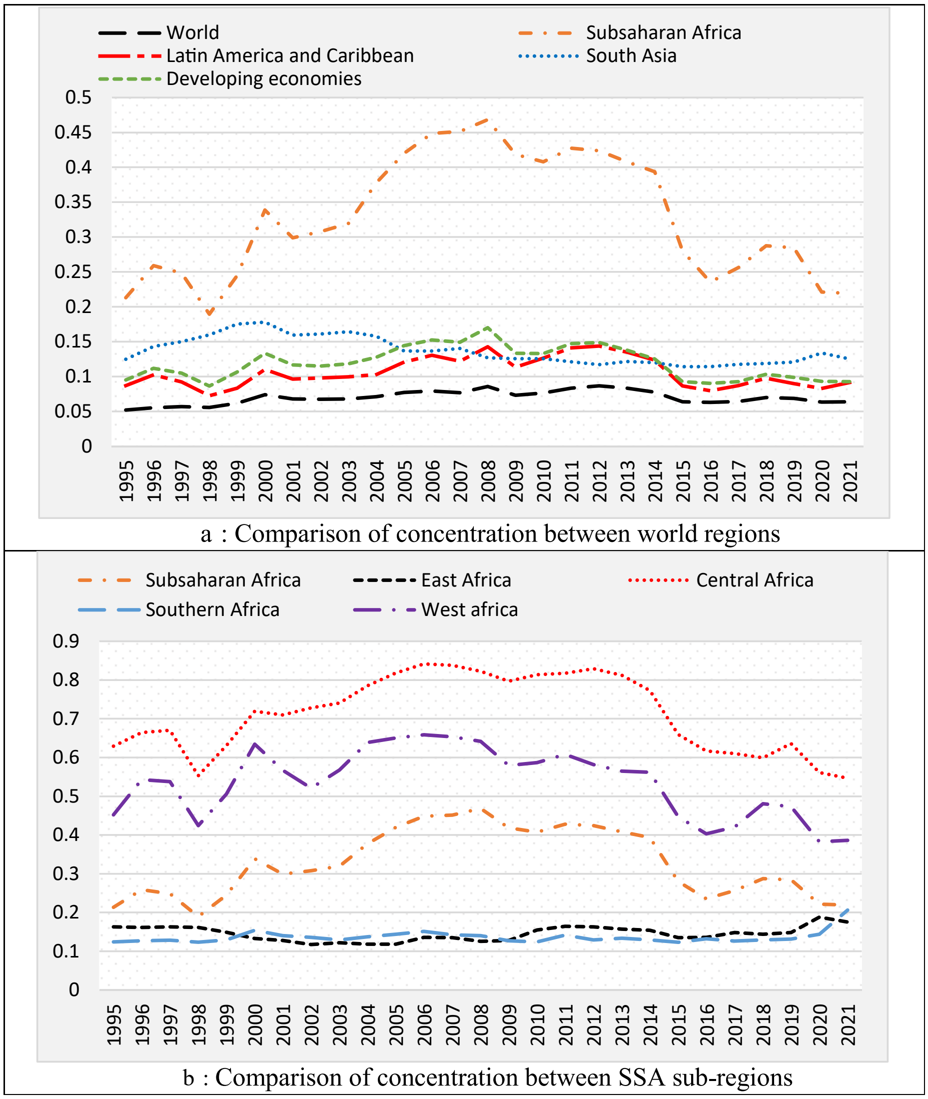

Fig. 1. Concentration index.

than, for instance, the Herfindahl-Hirschman index for developing countries prone to price changes such as those of Africa (Fosu, 2021; Fosu & Abass, 2019). As the Theil index is a concentration index, a relatively high level of the index corresponds to a concentration of exports, while conversely, a relatively low level of the index reveals a diversified export structure. Furthermore, the Theil index is a composite index and therefore a non-normalized index, so the formulation: one minus the Theil index (T) (D = 1 − T) cannot be applied (Drudi & Tassinari, 2014). Thus, a positive sign of the coefficient of the Theil index reflects a negative effect of increased diversification and vice versa. The Theil squared index (THEILSQUARE) is used in the estimation to verify the existence of a nonlinear relationship as suggested by Imbs and Wacziarg (2003).

3.3. Estimation technique

To estimate the nature of the relationship between export diversification and economic growth, we use a panel ARDL model based on the PMG estimator. Developed by Pesaran et al. (1999), the PMG estimator distinguishes between the short term and the long term while considering country heterogeneity. As such, it imposes an equality constraint on the long-run coefficients due to similarities in economic structure, technological catch-up, or the absence of arbitrage (Chudik & Pesaran, 2015), but considers that the model constant, short-run coefficients, and error variances may differ between individuals. This method also solves endogeneity problems in dynamic specifications by considering all explanatory variables as endogenous (Odugbesan & Rjoub, 2019). The empirical specification of the PMG estimator is as follows:

$$\Delta Y_{it} = \theta_i (Y_{i,t-1} - \beta' X_{it-1}) + \sum_{j=1}^{p-1} \Upsilon_{ij} \Delta Y_{i,t-j} + \sum_{j=1}^{q-1} \Upsilon'_{ij} \Delta X_{i,t-j} + \varphi_i + \xi_{it} \tag{2}$$

Where θ_i is the error correction speed of the adjustment parameter, X_{it-1} are explanatory variables, β' is the estimated long-term parameter, Υ_{ij} are parameters p to be estimated, Υ'_{ij} are parameters q to be estimated, φ_i represents individual fixed effects and ξ_{it} are error terms, $i = 1, \ldots, N$, and $t = 1, \ldots, T$.

The PMG estimator requires two preconditions. The first is that all variables are stationary at the level or first difference, and the second is that there is a long-run relationship between the variables. In this work, we apply three-unit root tests on panel data, namely the LLC test by Levin, Lin, and Chu (2002), the IPS test by Im et al. (2003), and the CIPS (Cross-sectional Im, Pesaran & Shin) test by Pesaran (2007). The LLC test tests the null hypothesis of the presence of unit roots and the alternative hypothesis of stationarity of the entire panel, while the IPS test tests the null hypothesis that each time series contains a unit root and the alternative hypothesis that only part of the series is stationary. The CIPS test is based on the same principle as the IPS test, the only difference being that it considers the cross-sectional dependence of the observations.

To test for the existence of a long-term relationship, the tests of Kao (1999) and Pedroni (2004) are performed. The tests of Kao (1999) are tests of the null hypothesis of the absence of cointegration of the Dickey–Fuller and Augmented Dickey–Fuller types, but which assume homogeneous cointegration vectors between individuals. The Pedroni tests (2004), in contrast to the Kao tests, take account of individual heterogeneity through individual parameters. Under the null hypothesis, they test for the absence of cointegration; under the alternative hypothesis, they admit the existence of a cointegrating relationship for everyone, with parameters that may differ from one individual to another.

After assessing the shape of the relationship between export diversification and economic growth, if there is a hump-shaped relationship, then the turning point (TP) of the curve is estimated as follows:

$$TP = \frac{\partial GDP_{it}}{\partial THEIL_{it}} = \alpha_1 + 2\alpha_2 THEIL_{it} = 0 \tag{3}$$

$$\text{Thus}, TP = -\alpha_1 / 2\alpha_2 \tag{4}$$

with α_1 being the estimated coefficient of THEIL and α_2 the estimated coefficient of the THEILSQUARE quadratic term.

4. Results and discussion

4.1. Preliminary tests

Table 1 presents the descriptive statistics and data sources. Looking at these variables, we see that with a mean value of *\$67182.51*, GDP has a standard deviation of *145,575.1* in SSA. The human capital index shows an average of *1.734*, with a less dispersed distribution across the region. The averages for employment, physical capital, and population growth rate are *7.637* million, *245,714.4* thousand, and *2.517%*, respectively.

Table 2 shows the results of the stationarity tests. According to the LLC test, the Theil index, Theil squared, the rate of depreciation of physical capital, and the rate of population growth are stationary at level, except for GDP, employment, the human capital index, and the stock of physical capital, which are stationary at first difference. The IPS test shows the same results, the only difference being that the rate of depreciation of physical capital is no longer stationary at level but rather at first difference. As for the CIPS test, it shows that under the hypothesis of cross-sectional dependence, only the

Table 1. Descriptive statistics.

Variables	Observations	Averages	Standard deviation	Minimum	Maximum
GDP	900	67,182.51	145,575.1	591.234	1,006,237
THEIL	900	3.240	0.863	1.228	5.406
THEILSQUARE	900	11.245	5.837	1.510	29.228
EMPLOYMENT	900	7.637	10.555	0.218	73.020
HUCAP	900	1.734	0.427	1.049	2.938
PHYCAP	900	245,714.4	576,770.6	5068.762	3,115,441
DELTA	900	0.044	0.010	0.017	0.083
DEMO	900	2.517	1.301	−16.880	16.225

Source: Authors.

Table 2. Stationarity test results.

Variables	LLC		IPS		CIPS	
	I(0)	I(1)	I(0)	I(1)	I(0)	I(1)
GDP	No	Yes	No	Yes	No	Yes
THEIL	Yes	–	Yes	–	Yes	–
THEILSQUARE	Yes	–	Yes	–	Yes	–
EMPLOYMENT	No	Yes	No	Yes	No	Yes
HUCAP	No	Yes	No	Yes	Yes	–
PHYCAP	No	Yes	No	Yes	No	Yes
DELTA	Yes	–	No	Yes	No	Yes
DEMO	Yes	–	Yes	–	No	Yes

Source: Authors' estimates.

Theil index, Theil squared, and the rate of depreciation of physical capital are stationary at level, while the rest of the variables are stationary at first difference.

The results of the cointegration tests reported in Table 3 show that the Kao test cannot accept the null hypothesis of no cointegration at the 1% threshold for all statistics. The seven Pedroni test statistics also fail to accept the null hypothesis of no cointegration at the 1% and 5% thresholds (for the fifth statistic). The results obtained from the two tests show that there is a long-term relationship between the different variables under consideration.

4.2. Relationship between export diversification and economic growth

From the estimation results reported in Table 4, we see that the coefficients of the error correction term (ECT) for all three estimations are significantly negative. This implies that the variables converge toward a long-term equilibrium in the event of distortion. Furthermore, we observe that the signs and significances of the coefficients of the sub-sample of countries with high export diversification are the same as those of the total sample; but contrary to those of the sub-sample of countries with low export diversification. This reveals that analyzing the relationship between export diversification and economic growth in SSA by considering all countries as a single sample may hide some important information about disparities between countries. Thus, the results that emerge at the global level are more a reflection of the effect of diversification on growth in countries with highly diversified exports.

Analysis of the results shows that the coefficient of the Theil index for the (E1) is significantly negative at 1%, i.e., an increase in export diversification leads to economic growth. In other words, a 1-point improvement in diversification translates into an increase in GDP of 0.397 thousand dollars for the total sample and 0.597 thousand dollars for the sub-sample of highly diversified countries. At the same time, we note a positive sign for the coefficient of the Theil quadratic variable, which is opposite to that of the Theil index. This means that for (E1), the relationship between export diversification and economic growth has an inverted U shape. In fact, the critical point is 3.4. This result corroborates Imbs & Wacziarg's (2003) seminal idea that the more diversified an economy is, the more it will tend to reach a high-income level, at which point it will tend to concentrate again. This result is also in line with Klinger and Lederman (2006), as well as Cadot et al. (2011) in developed countries, and Yuni et al. (2020) who, by classifying SSA countries according to their average income level, demonstrated the existence of a nonlinear relationship.

As for the sub-sample (E2), the export diversification coefficient reveals a negative effect. In other words, in the least diversified sub-regions, concentration is beneficial to economic growth. This result is like that found in the work of Gutierrez De Piñeres and Ferrantino (1997), Matthee et al. (2016), McIntyre et al. (2018), Lee and Zhang (2019), and Siswana and Phiri (2020). According to these authors, in countries with low revenue levels (export diversification), it is product specialization that drives economic growth. Indeed, for highly specialized economies, export diversification entails significant costs (Cherif & Hasanov, 2019). These costs prevent them from being efficient in all sectors of activity. It is therefore easier for these economies to remain specialized. However, unlike these studies, which fail to establish inflection points or thresholds in export-growth diversification regressions (Siswana & Phiri, 2020), the Theil squared coefficient in the present study clearly shows that at

Table 3. Cointegration test results.

Kao cointegration test		
	Statistics	*p*-value
Modified Dickey–Fuller test	3.0818	0.0010
Dickey–Fuller test	2.8168	0.0024
Augmented Dickey–Fuller test	2.1311	0.0165
Unadjusted modified Dickey–Fuller test	3.7721	0.0001
Unadjusted Dickey–Fuller test	3.7714	0.0001
Pedroni cointegration test		
Modified Phillips–Perron test	4.2044	0.0000
Phillips–Perron test	−14.2068	0.0000
Augmented Dickey–Fuller test	−13.6536	0.0000
Modified variance ratio	−5.7610	0.0000
Modified Phillips–Perron test	2.3252	0.0100
Phillips–Perron test	−12.2900	0.0000
Augmented Dickey–Fuller test	−12.0428	0.0000

Source: Authors' estimates.

Table 4. Results of the econometric analysis.

Variables	Coefficients		
	Total sample	Strong diversification (E1)	Low diversification (E2)
THEIL	−.397[a] (0.081)	−0.597[a] (0.121)	2.213[a] (0.557)
THEILSQUARE	0.057[a] (0.013)	0.088[a] (0.020)	−0.370[a] (0.088)
EMPLOYMENT	0.036[a] (0.004)	0.044[a] (0.05)	−0.070[b] (0.028)
HUCAP	1.933[a] (0.068)	2.022[a] (0.124)	−0.768[b] (0.337)
PHYCAP	0.269[a] (0.048)	0.046[a] (0.062)	0.280[b] (0.111)
DELTA	−0.377[a] (0.049)	−0.567[a] (0.094)	1.259[a] (0.208)
DEMO	0.074[a] (0.016)	0.079[a] (0.025)	0.103[a] (0.036)
ECT	−0.196[a] (0.048)	−0.185[b] (0.083)	−0.078[a] (0.033)
ΔTHEIL	−1.732 (1.997)	−0.132 (0.146)	−3.330 (3.841)
ΔTHEILSQUARE	0.274 (0.304)	0.039 (0.036)	0.510 (0.587)
ΔEMPLOYMENT	−0.101 (0.275)	−0.343 (0.493)	−0.071 (0.228)
ΔHUCAP	3.982[c] (2.286)	0.275 (0.996)	4.973 (5.917)
ΔPHYCAP	1.034[b] (0.305)	0.887[b] (0.282)	0.801 (0.477)
ΔDELTA	−0.082 (0.137)	0.242[b] (0.114)	−0.083 (0.170)
ΔDEMO	0.028 (0.247)	0.015 (0.022)	−0.011 (0.018)
CONSTANT	1.097[a] (0.271)	0.958[b] (0.434)	0.739[b] (0.331)
TURNING POINT		3.4	3
OBSERVATIONS	864	408	456
NUMBER OF i	36	17	19
NUMBER OF t	24	24	24

Source: Authors' estimates. ECT, error correction term.
Significance:
[a] 10%.
[b] 5%.
[c] 1%.

some point the relationship will reverse. Moreover, the results in Table 4 show that, from the minimum TP equal to 3, these specialized sub-regions will have to think about diversifying their economies. It therefore appears that, in the more concentrated sub-regions, the relationship between export diversification and economic growth is U-shaped.

About the control variables, the comparative analysis highlights that employment level and human capital are significantly positive for sub-sample E1 but are significantly negative for sub-sample E2. This reflects the fact that employment levels and human capital contribute to economic growth for sub-sample E1. This can be explained by the job creation resulting from the diffusion effects of the technological spin-offs of export diversification. On the other hand, the negative effect of these variables could be explained by the fact that poorly diversified economies are more oriented toward capital-intensive mining activities and do not allow the rural poor to benefit from the spillover effects of resource rents (whether in terms of job creation or skills enhancement characterizing any modern economy) (López-Cálix, 2020).

Physical capital has a positive effect on growth in the different samples. This result is not surprising, as it remains in line with endogenous growth theories. On the one hand, the latter postulates that domestic investment helps stimulate economic growth (Napo & Adjande, 2019). On the other hand, it is argued that the opening of international capital markets generates abundant capital flows to countries with reduced capital, accelerating convergence in poor countries (Napo & Adjande, 2019). Population growth positively affects economic growth in all three samples. This result follows from the work of Al-Marhubi (2000) and could be explained by the fact that in SSA, the population primarily represents an abundant and cheap labor force. Furthermore, in the African context, the most developed activities are those in the agricultural and informal sectors, which do not always require a high level of skilled human capital. Finally, population growth also means an increase in the size of countries' domestic markets, which in turn means higher revenues for companies due to economies of scale.

5. Conclusion

This paper explores the existence of a mixed relationship between export diversification and economic growth in SSA. It leads to three key observations: first, analyzing the relationship between export diversification and economic growth in SSA by considering all countries as a single sample hides some important information about disparities between countries and reflects the absorption of effects in less diversified countries. Second, export diversification has a positive effect on economic growth when countries are more diversified and a negative effect when they are less diversified. And third, the relationship between export diversification and economic growth is nonlinear. It is, therefore, suggested that export diversification policies be implemented in a way that considers the specific characteristics of each economy. Less diversified countries could, for example, maximize the acquisition of raw material rents during their period of concentration. These rents will later be used to bear the costs (infrastructure, workforce training, technology, etc.) associated with economic diversification. As they diversify, they will need to identify priority sectors and implement industrialization policies in line with their comparative advantages (Lin & Monga, 2010). As for the more diversified countries, to further benefit from their diversification, they could accelerate their integration into the African Continental Free Trade Area (ACFTA). Unlike other zones, the ACFTA offers greater openness to new markets, and hence improved competitiveness, by removing tariff and non-tariff barriers and improving the regulatory framework.

Ethics Information

None.

Author Contribution

The paper was prepared by all authors in the equal proportion.

Conflict of interest

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Appendix 1.

Low diversification (E1)		Strong diversification (E2)	
Southern Africa	East Africa	Central Africa	West Africa
Botswana	Burundi	Angola	Benin
Eswatini	Ethiopia	Cameroon	Burkina Faso
Lesotho	Kenya	Central African Rep.	Ivory Coast
Namibia	Madagascar	Congo Republic	Gambia
South Africa	Malawi	Congo Dem. Rep.	Ghana
	Mauritius	Gabon	Liberia
	Mozambique		Mali
	Rwanda		Mauritania
	Tanzania		Niger
	Uganda		Nigeria
	Zambia		Senegal
	Zimbabwe		Sierra Leone
			Togo

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